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Unit – 5 : Session – 3 : SLO – 1 | Graph Traversal — Depth First Search (DFS)

■ I. Fill in the Blanks

- DFS stands for DEPTH FIRST SEARCH.
- DFS uses STACK data structure for its implementation.
- In DFS, a vertex is marked as VISITED when it has been visited.
- The time complexity of DFS for a graph with V vertices and E edges is $O(V + E)$.
- DFS traversal can be implemented using both RECURSIVE and ITERATIVE methods.

■ II. True or False

- DFS always visits the shallowest unvisited node first — FALSE (that is BFS).
- DFS can be used to detect cycles in a graph — TRUE.
- DFS is suitable for finding the shortest path in an unweighted graph — FALSE (BFS is).
- The recursive implementation of DFS uses system call stack — TRUE.

■ III. Short Answer Questions

- DFS: a traversal that starts at a vertex and explores as deep as possible along each branch before backtracking, using a stack (or recursion).
- Applications: cycle detection; topological sorting; also maze solving and finding connected components.
- Data structure: STACK (explicit or the system call stack in recursion).
- DFS vs BFS: DFS explores depth-first using a stack; BFS explores level-by-level using a queue and finds the shortest path in unweighted graphs.

■ IV. Solve and Write — Given Undirected Graph (A-H)

- Adjacency list:
 - A -> B, C B -> A, D, E C -> A, F, G D -> B, H
 - E -> B, H F -> C, H G -> C, H H -> D, E, F, G
- DFS traversal from A (neighbours in alphabetical order): A, B, D, H, E, F, C, G
- In recursive DFS, the system internally uses the CALL STACK (system stack).